whole endeavor of studying the cosmic end times. “The study of the remote future still seems to be as disreputable today as the study of the remote past was thirty years ago,” he wrote, pointing out the scarcity of serious papers approaching the subject.[fn2] He continued with a cosmological call to arms: “If our analysis of the long-range future leads us to raise questions related to the ultimate meaning and purpose of life, then let us examine these questions boldly and without embarrassment.”

I can’t exactly say that cosmic eschatology has, after all this time, finally received its proper level of respect as an academic discipline. It’s still rather rare to find papers in the physics literature that examine our ultimate fate with the same rigor and depth as they do our origins. But studies of both ends of the timeline help us, in different ways, to examine the principles of our physical theories. Beyond the insight they might provide into our future or past, they can help us understand the fundamental nature of reality itself.

“By thinking about the end of the universe, just like with its beginning, you can sharpen your own thinking about what you think is happening now, and how to extrapolate. I feel like extrapolations in fundamental physics are essential,” says Hiranya Peiris, a cosmologist at University College London. In 2003, she led one of the teams interpreting the first detailed view of the cosmic microwave background with the Wilkinson Microwave Anisotropy Probe (WMAP) satellite and she has since then maintained a position at the leading edge of observational cosmology. In recent years, she’s set her sights on using observational data, simulations, and tabletop analogs to test some of the key elements of early- and late-universe physics like the creation of “bubble universes” in cosmic inflation and the mechanics behind vacuum decay. In studying all these questions, her motivation is the same. “I know this period needs to be understood. How what we’re doing now will map directly onto those periods is still not clear, but I think we’ll learn something about fundamental theory by doing this work.”

We certainly have a lot to learn. Cosmology and particle physics are in an awkward position at the moment; both have, in some ways, been victims of their own success. In each field, we have a very precise and comprehensive description of the world that works extremely well in the sense that nothing has been found to contradict it. The downside is that we have no idea *why* it works.